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Artificial intelligence of things based approach for anomaly detection in rotating machines☆

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ABSTRACT

In the present era of Industry 4.0, Artificial Intelligence (AI) and Internet of Things (IoT) are revitalizing the predictive maintenance systems for industrial rotating machines through real-time approaches. In this work, an Artificial Intelligence of Things (AIoT) based framework is developed and implemented for anomaly detection in rotating machines through vibration monitoring. Signals are captured using a low-cost MPU6050 accelerometer with Raspberry Pi 4B as an edge device. The edge-centric mechanism is based on Support Vector Machine (SVM) for effective implementation on the edge device. A dedicated web-based platform is designed to communicate with the edge device for real-time decisions. The developed framework is validated on three rotating machines, including gear test rig, bearing test rig, and induction motor. The results demonstrate the robustness of the proposed method for anomaly detection on different machines. Also, the developed framework has the capability to remotely monitor and control the machines in real-time.

1. Introduction

Rapid automation in manufacturing industries involves a complex machinery loop system that demands an appropriate and real-time feedback system for effective maintenance [1]. Anomaly detection or fault detection in this regard become the most viable practice requiring real-time decision-making as the failure of machines could lead to catastrophic loss. The rapid development in information and internet-based technologies has become a boon for developing intelligent and real-time solutions for the predictive maintenance of rotating machines with remote monitoring [2]. The anomaly detection task has been practiced with a model-based approach for many years. In the model-based approach, a mathematical model and physical laws are used to simulate the physical properties and characteristics of the system to be monitored, such as mass, stiffness, and damping matrices [3]. However, these methods heavily rely on the knowledge of the machine's internal structure and other factors, such as loading and its nature. It involves complex mathematical expressions, and a bunch of such complex expressions could be tedious to handle for actual industrial systems. Moreover, this approach is inadequate for updating real-time detection with newly observed data. In the fault detection process, statistical methods found their place in recent decades with the strong support of probabilistic models [4]. However, the different stochastic processes involved in these methods further burden the computational process.

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The recent adoption of methodology in fault detection is data-driven approaches involving signal processing and AI [5]. Data-driven approaches in the Industrial Internet of Things (IIoT) assisted system are preferred over model-based and statistical approaches. These approaches provide enhanced interoperability through ontologies to provide high-level reasoning-based inquiry responses to non-expert users [6]. Among these, AI-based methods achieved universality for the fault detection task and other engineering applications. These methods of anomaly detection majorly involve data acquisition, feature extraction, and then classification through various AI-based methods, including Machine Learning (ML) and Deep Learning (DL) [7]. These methods have achieved excellent results in the tasks related to anomaly detection. Furthermore, DL-based algorithms are getting much attention with the rapid leveraging of computational power. ML-based methods, however, find effective implementation on edge devices with proper feature extraction due to their low consumption of computational power. However, the implementation of these algorithms on edge devices, along with the provision of remote monitoring to form a cost-effective fault detection system, is challenging. In the proposed work, an AIoT-based solution has been developed for the real-time detection of anomalies in different rotating machines by implementing the machine learning model on an economical edge device and providing remote monitoring and control. The effectiveness of the proposed idea is validated through experiments on three different rotating machines.

The remaining parts of the paper have the following sections: Section 2 elaborates on the related work, Section 3 describes the proposed AIoT-based methodology, Section 4 explains hardware and data acquisition, Section 5 explains the obtained results and related discussion, Section 6 contains managerial and social implication, followed by Section 7 having the conclusion.

2. Related work

The above-discussed practices heavily rely on expert knowledge and primarily involve in-house monitoring [8]. Further, in harsh working conditions, such practices are quite challenging and need a system that can remotely work with significant reliability and robustness. In such a scenario, IoT establishes its presence for monitoring and real-time anomaly detection of the machines. The IoT communicates assertive information through wireless sensors and transmission technologies such as Wi-Fi, Bluetooth, ZigBee, WiMax, etc. The IoT provides innumerable advantages to humanity by creating an ecosystem providing smart work without the bounds of place and time [9]. IoT for condition-based maintenance becomes crucial to make the maintenance system smart enough to handle a large amount of data effectively and efficiently [10]. The advancement of wireless technology has made real-time data access simple by integrating hardware and software architecture. Researchers have done tasks to collect the data through different embedded devices with various sensors, followed by a flow of this data to the cloud. Four sensors were used for condition monitoring of a three-phase induction motor, and SVM with cloud computing was proposed. The applicability of reduced features for sending to the cloud was also demonstrated [11]. An Arduino-based microcontroller was used to collect and transfer the data through Wi-Fi to monitor a wind turbine prototype using heterogeneous sensors. Three ML methodologies were used for analysis using the local machine [12]. Another research work demonstrated the study related to abnormal conditions in the motor, including motor-load, uncoupling, and misalignment using speed and rotor position information. IEC61784–2 module-based real-time ethernet network was defined, and the process field network protocol that belongs to the application layer was used for information gathering. The accuracy obtained was quite appreciable, in the range of 87.5 %–100 % [13].

The industrial system structure is usually complex and involves multiple rotating machines generating massive data. The flow of this data to the cloud and its handling creates many issues involving its compression, security, etc. [14]. Recently, the integration of AI with IoT has been getting much attention for designing industrial systems that are intelligent and smart. In such a system, data handling is done at the edge, and the decision is sent to the cloud with further control. Such a system can be enumerated as an AIoT-enabled system that more intelligently examines and counter external promotion [15]. In the case of cloud-centric IoT, it is infeasible to process large amounts of data because of the destructive impact of the volume in the cloud [16]. Although the cloud center is usually equipped with massive computational capacity, such a system is ill-suited in the context of time-criticality, and applications require privacy-sensitiveness due to the enormous burden on the network's bandwidth. In continuous monitoring of the rotating machines, a large amount of data is generated and needs to be stored for processing and analysis. So, sending this data to the cloud and processing that, and then making a decision could be tedious and involve the risk of triggering the fault in the meantime.

Further, other challenges include in herent latency limitations related to network communication and the chances to disclose sensitive and critical information in data offloading from the cloud [17]. This critical issue can be adequately resolved through the integrated application of edge computing, AI, and IoT. Edge is about processing data at a faster rate and in a massive volume close to the generation point, providing action-driven solutions in real-time. The data is acquired through edge devices that might include some microcontrollers enabled with embedded sensors, single-board computers, etc. This involves data acquisition using Micro Electro Mechanical System (MEMS) based low-cost sensors, data preprocessing, and feature extraction to get meaningful patterns. The data storage at the edge ensures protection from the threat of cyber-attacks, a light access burden, and a lesser transmission delay. Further, this enables the user to analyze the data in case of the unavailability of the internet on the user side. Finally, the classification and detection of anomalies can be done through the edge devices and sending the real-time decision to the cloud, and subsequent actions can be made to avoid significant losses [18]. In the present work, edge-computing powered AIoT-based architecture is implemented for the fault detection of three rotating machines. The system is designed to make decisions at the edge, and its mutual communication with the web-based platform for remote monitoring. The following are the main contributions of this work.

(1) A dedicated AIoT-based framework is designed and deployed for the real-time anomaly detection of rotating machines.
(2) Validating the developed system on three rotating machines, including a gear test rig, bearing test rig, and induction motor.
(3) Real-time anomaly detection and health monitoring through the developed AIoT-based system.

3. Proposed methodology

The proposed methodology is based on the machine learning implementation on edge device and subsequent communication and control of different machines through a dedicated web-based AIoT platform. The complete flow diagram of the AIoT-based methodology is shown in Fig. 1.

A total of four prevalent classification models are initially trained and tested at the local machine, and their performance is evaluated. The training is done through the extracted features from the vibration data. Then, the substantially performing classification model is deployed on the edge device for the real-time fault detection system.

3.1. Feature extraction and classifier training

For the training of different classifiers, nine statistical features are extracted, including mean, Root Mean Square (RMS), minimum value, maximum value, standard deviation, kurtosis, skewness, peak-to-peak value, and energy. These features are used for real-time anomaly detection and health monitoring of different rotating machines. Details of these features are given in Table 1.

The extraction of features involves a sliding window-based strategy. The expression for the number of extracted windows from the acquired vibration signal is given here:

$$p = \lfloor \varnothing \rfloor + f(\varnothing) \tag{1}$$

Here, $\varnothing = \frac{W-s}{(1-\alpha).s}$, $\lfloor \varnothing \rfloor$ denotes its floor value, W denotes the length of signal, s is window size, and $\alpha$is the overlapping parameter. Here

$$f(\varnothing) = \begin{cases} 1 & if \ \lfloor \varnothing \rfloor = \varnothing \\ 2 & if \ \lfloor \varnothing \rfloor < \varnothing < \lfloor \varnothing + 1 \rfloor \end{cases} \tag{2}$$

The total number of data points (i.e. length of the signal) in the signal is 40,000, an overlap of 75% is taken i.e. $\alpha = 0.75$, and window size is taken as 28, which gives a total of 5710 windows. Features are extracted from each window resulting in 5710 values for each feature. Here, nine features are taken, so the final feature set becomes 5710×9. After extracting these features, they are normalized from -1 to 1 to avoid the knock-on effect and better facilitation of learning.

Four classification models are then used for training and testing, including Gaussian Naïve Bayes (NB), k-Nearest neighbor (k-NN), Recurrent Neural Network (RNN), and SVM. NB is a prevalent supervised ML model considering the probabilistic approach. It operates under the assumption of naïve independence between feature vector elements. The probability distribution of the classes under consideration is computed. Mathematically it is given as:

$$\lambda_k = \underset{k \in 1,\ldots,K}{\operatorname{argmax}} P(\lambda_k) \prod_{j=1}^{n} p\left(x_j / \lambda_k\right) \tag{3}$$

Here $P(\lambda_k)$ is the prior probability of λ_k class with $p(x_j/\lambda_k)$ is its likelihood, n is the number of elements in the feature vector, and K is the number of classes. NB is supposed to work well on high-dimensional data. More details related to NB can be found in the article

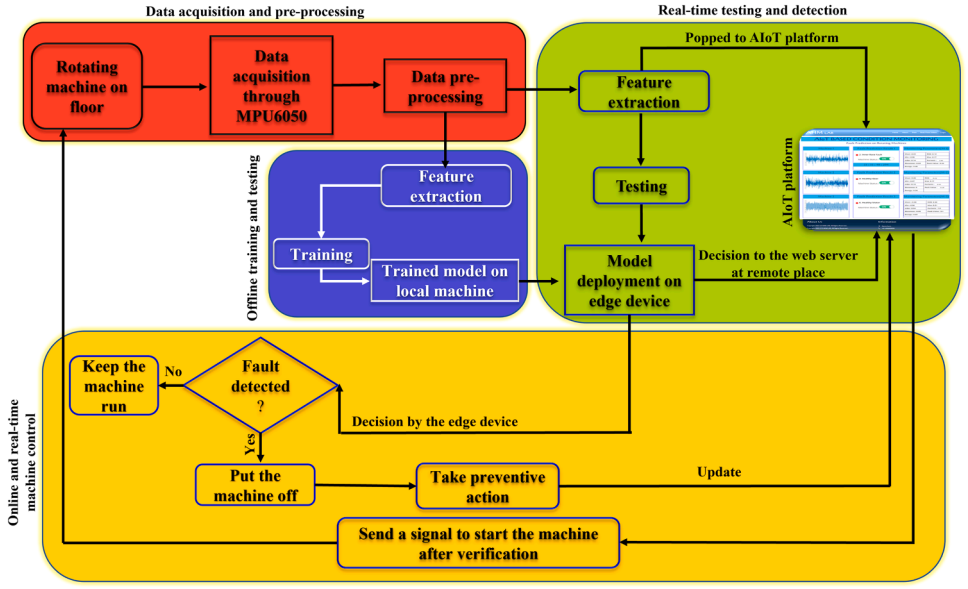


Fig. 1. Working methodology of AIoT-based anomaly detection of rotating machines.

Table 1
Mathematical expressions of extracted features.

Feature	Expression	Feature	Expression
Mean (x_{mean})	$\frac{1}{N}\sum_{k=1}^{N}(x_k)$	RMS (x_{rms})	$\sqrt{\frac{1}{N}(\sum_{k=1}^{N}(x_k)^2)}$
Minimum value	$\min(x_k)$	Maximum value	$\max(x_k)$
Standard deviation (x_σ)	$\sqrt{\frac{1}{N}(\sum_{k=1}^{N}(x_k - x_{mean})^2)}$	Kurtosis	$\frac{\sum_{k=1}^{N}(x_k - x_{mean})^4}{N.(x_\sigma)^4}$
Skewness	$\frac{\sum_{k=1}^{N}(x_k - x_{mean})^3}{N.(x_\sigma)^3}$	Peak-to-peak value	$\max(x_k)$-$\min(x_k)$
Energy	$\sqrt{\sum_{k=1}^{N}\lvert x_k\rvert^2}$		

[19]. Another prevalently used non-parametric supervised classifier is k-NN. In the present work, weighted k-NN is used with an improvement in the traditional k-NN method with the significant advantage that the recognition accuracy is less affected by the neighborhood size, i.e., k. More details of weighted k-NN can be found in the article [20]. Over the last decade, leveraged computational power has attracted researchers to use DL models for various engineering applications. Among the available DL models, RNN is supposed to work excellently on time series data. RNNs constitute a powerful class of computational models capable of learning arbitrary dynamics. It combines knowledge from memories with current information to make a prediction. The significant distinction between RNN and traditional neural networks lies as in RNN, neurons use the role of the preceding neuron. If the input feature vector is x_n and ψ_n as the actual output, then a hidden state $h_{\ln}$ is produced and is given as:

$$h_{\ln=}\sigma_h(W_{xh}x_n + W_{hh}h_i + b_h) \tag{4}$$

Here, W_{xh} is utilized for the linear transformation of input and is known as kernel weight matrix, W_{hh} is known as the recurrent kernel weight matrix, and is used for the recurrent state (h_i) linear transformation. σ_h is termed as Rectified Linear Unit (ReLU). A fully connected or dense layer is then followed for the prediction or classification of the output and is given as:

$$\widehat{\psi}_n = \sigma_\psi(W_{h\nu}h_{\ln} + b_\nu) \tag{5}$$

Here, σ_ψ could be a softmax or sigmoid activation function. In the present case, the softmax activation function is used. $W_{h\nu}$ is used for the linear transformation of the hidden state i.e., $h_{\ln}$ and b_ν is termed as a bias for the output layer.

Recently, SVM has gained much popularity among researchers for various engineering applications. This is because SVM has numerous advantages, such as it is memory efficient and supposed to be effective in high dimensional space. SVM works on the principle of the hyperplane, and finding an optimized hyperplane is effective in classifying the data correctly between different classes. SVM operates on convex optimization, which is effective as it is assured of optimality in the output, resulting in a global minimum and not the local minimum. SVM is also less affected by outliers, which is more critical when dealing with data related to fault detection with low-cost sensors. Mathematically the expression of hyperplane is given as:

$$y(x) = \theta^T\phi(x) + b \tag{6}$$

Here, θ is termed as an orthogonal vector and ϕ is the nonlinearity function or kernel in another term. With the help of a kernel, data can be efficiently made separable using the higher dimensional space. The feature vector closest to the hyperplane (support vectors) influences the final classification result. The final classification is obtained through Lagrangian optimization algorithms and kernel and is given as:

$$f(x) = sign\left(\sum_{i=1}^{N}\alpha_i y_i\phi(x, x_i) + b\right) \tag{7}$$

Here, α is termed as Lagrangian multiplier and $\phi(x, x_i)$ is a kernel function incorporated for high dimensional mapping. A radial basis function kernel is utilized for the current study and is given as:

$$\phi(x, x_i) = \exp\left(\frac{-\ ||x - x_i||^2}{2\sigma^2}\right) \tag{8}$$

Here, $||x - x_i||^2$ is termed as squared Euclidean distance and σ is a free parameter. Other important parameters for SVM training included the box constraint value, which is 1 in the present case. For all four classification algorithms, 20% of the data is used for testing purposes, i.e., 1142 samples of each class. In the case of RNN, 50 epochs are considered with a learning rate of 0.001, and the ReLu activation function is used. 'Adam' optimizer is used in the training of the RNN model.

The performance of the above-discussed classification algorithms is also evaluated in terms of six measures, including precision, recall, F1 score, specificity, Mathew Correlation Coefficient (MCC), and kappa value. These measures are calculated from various values extracted from the confusion matrix X. These values include True Positive (TP_X), False Positive (FP_X), True Negative (TN_X), and False Negative (FN_X). The F1-score is frequently applied among these parameters to quantify the classifier's performance. Specificity is computed as the number of correctly detected faults divided by the total number of faults that occurred. The MCC is a statistical rate

that is highly trusted because it yields a high score under the condition that predictions produce fair results in all the used classes, as evaluated in the confusion matrix. Cohen's kappa statistic is used to measure interrater reliability or interobserver agreement. These agreements are denoted as P_o (observed agreement) and P_e (expected agreement). It is an efficient parameter for describing the complete picture of a multi-class classification problem. The mathematical expressions of these performance measures are given in Table 2.

3.2. Development of AIoT platform

The AIoT platform is developed using a web development framework that eliminates the involvement of any third-party software. The data is stored using the database close to the edge server to predict different fault conditions. This end-edge-cloud collaborative approach lightens the data transmission burden. The interface development involves real-time data monitoring with the simultaneous values of nine statistical features (or health indicators) discussed earlier. The decision is perceived based on real-time testing through the deployed classification model, and the platform can be accessed from any part of the globe with the web address of the platform through proper login. Data acquisition continuously occurs with the simultaneous extraction of features and detection of the machine's health. The real-time decision the deployed model predicts is reflected on the platform's Graphical User Interface (GUI). The embedded edge device is coded to have the capability to turn off the machine whenever a faulty condition gets predicted and produce related status on the platform, as shown in Fig. 2.

Although only real-time data is tested, the complete data history is simultaneously stored in the database at the edge. This data storage at the edge ensures protection from the threat of cyber security, a light access burden, and a lesser transmission delay. Further, this enables the user to analyze the data in case of the unavailability of the internet on the user side. Once the necessary repair or change of component occurs, an update is sent to the user at a remote place, and the machine can resume operation after proper verification.

4. Hardware and data acquisition

For data acquisition and real-time anomaly detection, Raspberry Pi 4B is used as an edge device. Several factors make it a preferred choice, including affordability, reasonable performance, small size, low consumption of power, and support compatibility. Technical specifications of it include a 64-bit quad-core processor with Broadcom BCM2711, quad-core Cortex-A72 (ARM v8), 4 GB RAM, and a dual-band 2.4/5.0 GHz wireless network. Furthermore, the MPU6050 accelerometer is used for the acquisition of vibration data. The leading factors in adopting this MEMS-based accelerometer include its significantly lesser cost, low power consumption, and compatibility with digital interfacing. The operating voltage range of MPU6050 is 2.3–3.4 V, with a maximum consumption of 3.9 mA. It has a vibration measuring range from ± 2 g to ± 16 g with a precise calibration tolerance of 3%. Here, ± 4 g is selected as per the application related to rotating machines. MPU6050 is based on I2C interfacing, and the operating temperature range is -40 to 85 °C. A robust adhesive Beeswax (Procured from Bruel & Kjaer) with a skinny layer is used to mount the MPU6050, ensuring firm contact with the measuring surface. The proposed AIoT-based methodology is applied to the three different experimental setups of rotating machines at the laboratory level, including the gear test rig, bearing test rig, and induction motor. A relay module is used to trip the machine "ON" or "OFF" from a remote place.

4.1. Gear test rig (GTR)

As an essential component, gear is used in each industry for the alteration of mechanical speed in a reduced or increased form. It has a wide application, including aircraft, windmills, and large ships. Experimentation is done on a gear test rig run by a 3-phase, 440 V, 1 HP induction motor used as a prime mover and a single-stage bevel gearbox coupled with a flexible coupling. Variable Frequency Drive (VFD) is used to control the speed of the prime mover. A total of 20 teeth in the input gear that is coupled with the prime mover and gear ratio is 1.5:1. Here, binary conditions are considered, including healthy gear labeled as GHL and faulty gear labeled as GFG. In case of faulty gear, one of its teeth is worn out. The complete experimental setup for gear fault detection is shown in Fig. 3.

4.2. Bearing test rig (BTR)

The bearing test rig consists of proprietary bearing support that allows for the simulation of a variety of test scenarios under various conditions. Two more plummer block housings hold the bearing drive shaft, which is connected to a 220 V, 2 HP DC motor. This bearing test setup uses an SKF bearing (BB1B420205), and the complete setup of the bearing test rig is shown in Fig. 4. The inner and outer diameters of the bearing are 25, and 52 mm, respectively, and the pitch diameter is 37.9 mm. Here, a multi-class classification

Table 2
Mathematical expressions of performance measures.

$\text{Precision(P)} = \frac{TP_X}{TP_X+FP_X}$	$\text{Recall(R)} = \frac{TP_X}{TP_X+FN_X}$	$\text{F1} - \text{Score} = \frac{2PR}{P+R}$
$\text{Specificity} = \frac{TN_X}{TN_X+FP_X}$	$\text{MCC} = \frac{(TP_X.TN_X) - (FP_X.FN_X)}{\sqrt{(TP_X+FP_X)(TP_X+FN_X)(TN_X+FP_X)(TN_X+FN_X)}}$	$\text{KappaValue} = \frac{P_o - P_e}{1 - P_e}$

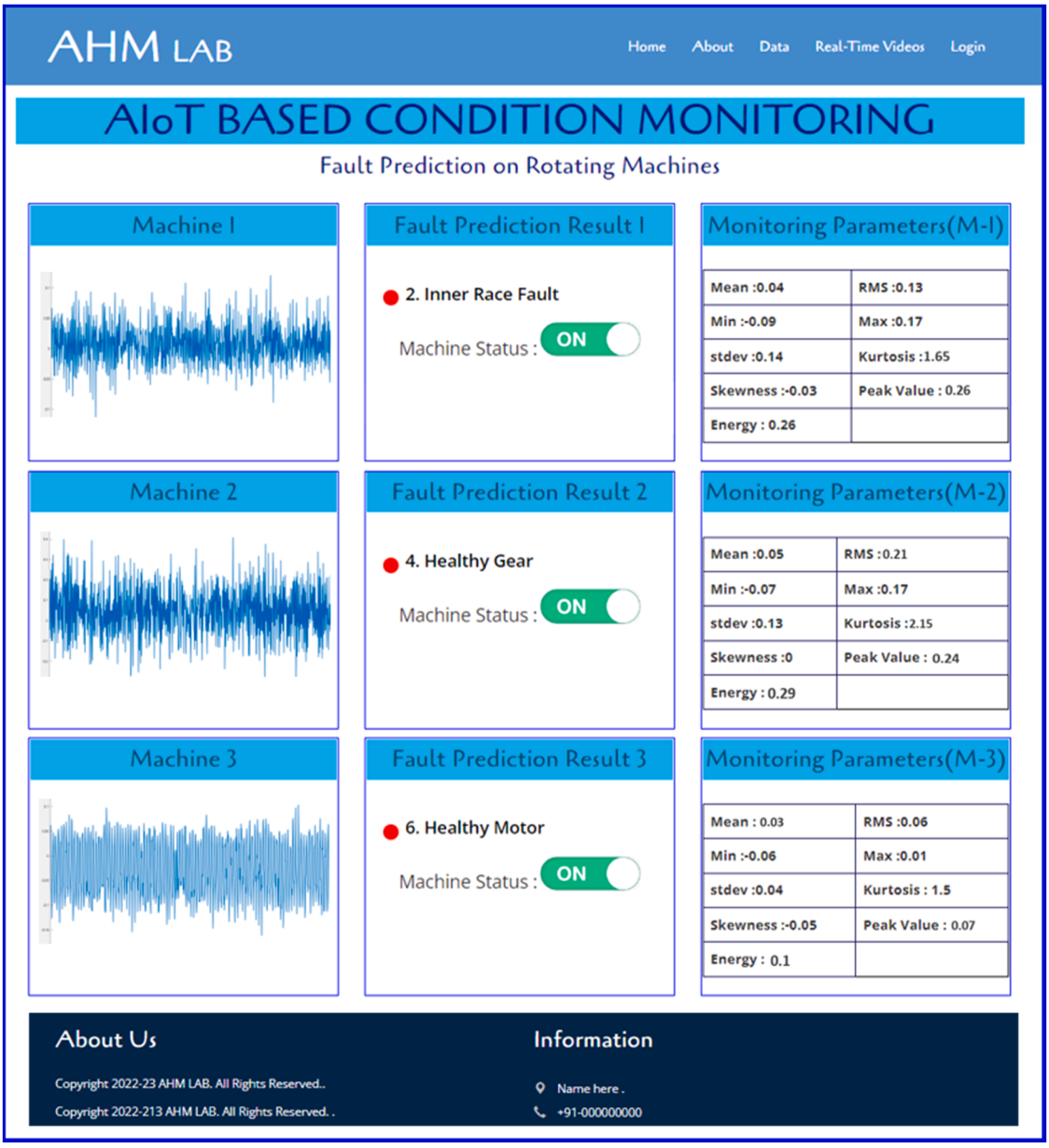


Fig. 2. AIoT-based platform.

problem is considered. A total of three conditions of the bearing are considered, including one healthy, inner race fault, and ball fault labeled as BHL, BIR, and BBF, respectively. BIR fault is created by electric discharge machining, making a through hole of 2 mm. BBF fault is created by making a 1 mm rub on one of the balls using a surface grinder.

4.3. Induction motor (IM)

An induction motor is preferred for the prime mover in most industries and is used to rotate small to large machines. The setup to experimentally simulate various induction motor faults is shown in Fig. 5. The motor specification is 3 phase, 440 V, 1 HP, and a VFD is used to control the motor's speed. An input voltage of single-phase of 50/60 Hz and 115/220 VAC is applied to give power to the VFD that controls the motor speed. Apart from a healthy motor (labeled as MHL), two faulty motors are considered, one has a fault in the bearing, and the other has a rotor misaligned and is labeled as MBR and MML, respectively. The MBR fault is generated by creating an outer and an inner race defect of a 1 mm groove on the bearing. Similarly, to create the MML fault, two jack bolts are adjusted to create an offset.

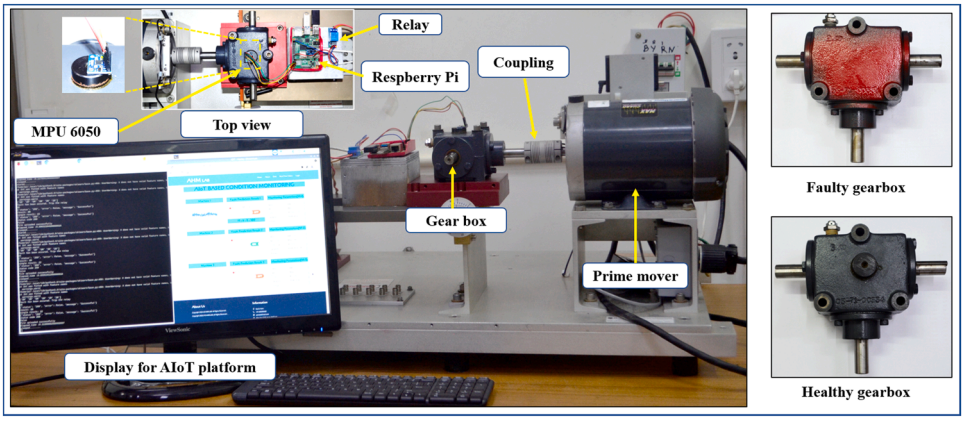


Fig. 3. Experimental setup for gear fault detection.

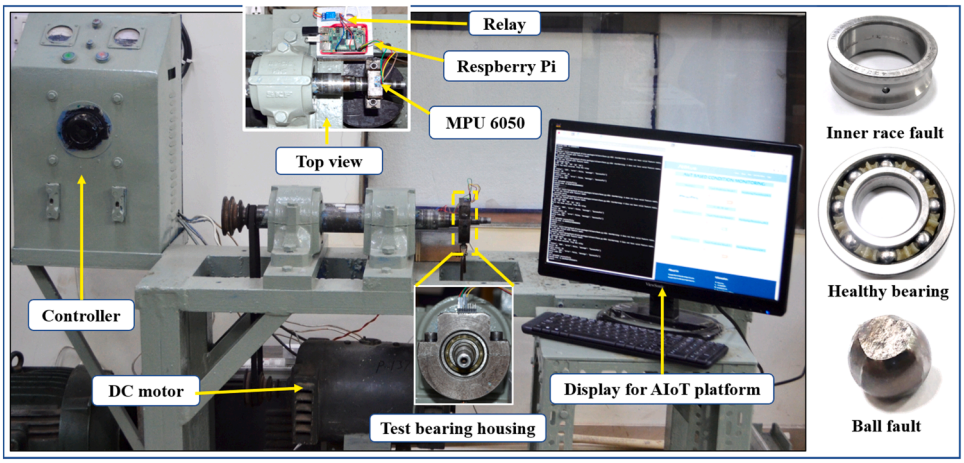


Fig. 4. Experimental setup for bearing fault detection.

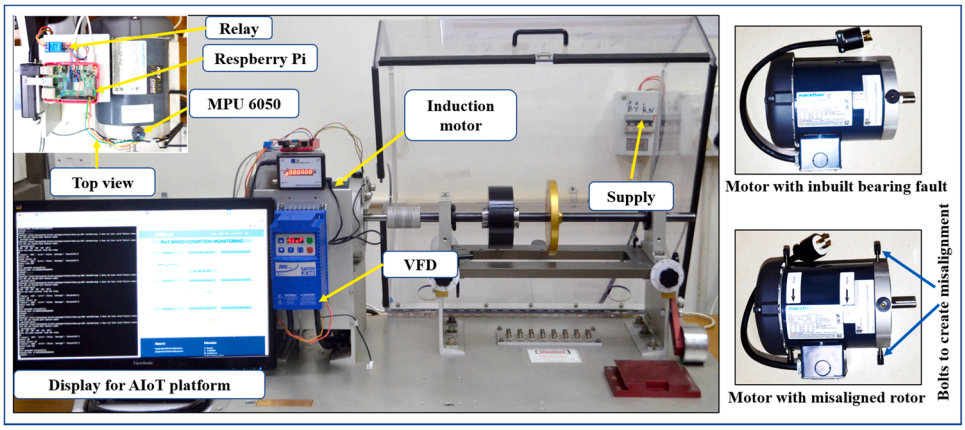


Fig. 5. Experimental setup for induction motor fault detection.

4.4. Data acquisition

For the acquisition of data, experimentation is done on all three machines at three different speeds, including 1140 rpm (19 Hz), 1380 rpm (23 Hz), and 1740 rpm (29 Hz). These speeds are selected to inhibit any overlap with the supply frequency, i.e., 50 Hz. MPU

6050 accelerometer is used to acquire vibration data in the radial direction. The radial direction mounting is done as most fault-related information can be captured by adequately placing the transducer in this direction. The accelerometer is calibrated before the data acquisition to ensure reliability in the measurement. Raspberry Pi and the relay module are firmly attached to a wooden board to ensure proper hardware placement and avoid any hindrance in data acquisition. Initially, the data for the healthy condition is captured in each case, followed by the faulty conditions. The data is acquired for 240 s in all conditions described earlier. The complete dataset for each condition is 40,000 data points of uniaxial vibration in the radial direction. Further, before starting the experiments, a pilot study is conducted to ensure the proper working of the hardware setup. The following are some assumptions that are made during the real-time conduction of experiments.

(1) There is no residual misalignment and unbalance in the bearing test rig during experimentation.
(2) The support bearings housings are rigid, enduring the radial and axial load.
(3) The pinion and gear meshing is assumed to be proper when capturing the data from the gear test rig.
(4) The prime mover for the bearing and gear test rig is in healthy condition.
(5) The effect of minor load fluctuations is neglected while capturing the data on misalignment and bearing faults in the induction motor.
(6) The internet connection is stable, and the AIoT platform has proper communication with the edge device.

5. Results and discussion

The methodology is validated on three rotating machines described in Section 3, i.e., GTR, BTR, and IM. The best-performing classification model is deployed on the edge device for real-time and remote-level detection among the four classification models. Other factors for the deployment are the amount of computational burden on the edge device and the offloading of real-time detection decisions on the AIoT-platform. This section follows the description and explanation of classifiers' performance in terms of classification accuracy for each machine. After that, the detailed results of the best-performing classifier are explained in terms of confusion matrices and respective six performance measures, as described earlier. Also, the feature visualization is given for a better understanding and unique insight into the functional capability towards fault detection. Further, the results related to overall system performance in real-time through a developed AIoT-based system are presented in the subsequent part. The classification results for the three rotating machines are as follows.

5.1. Classification results for the GTR dataset

In the case of gear fault detection, a binary classification problem is considered. In industries, worn-out gear teeth are a major problem. The visualization of the extracted features is provided in Fig. 6 at 19 Hz, 23 Hz, and 29 Hz. It can be observed that most of the features have a clear margin among the two classes in all the three-speed conditions that help the classifier to make a clear decision about the gear conditions.

The classification accuracy of all four classifiers is given in Table 3. The performance of the four considered classifiers is found to be

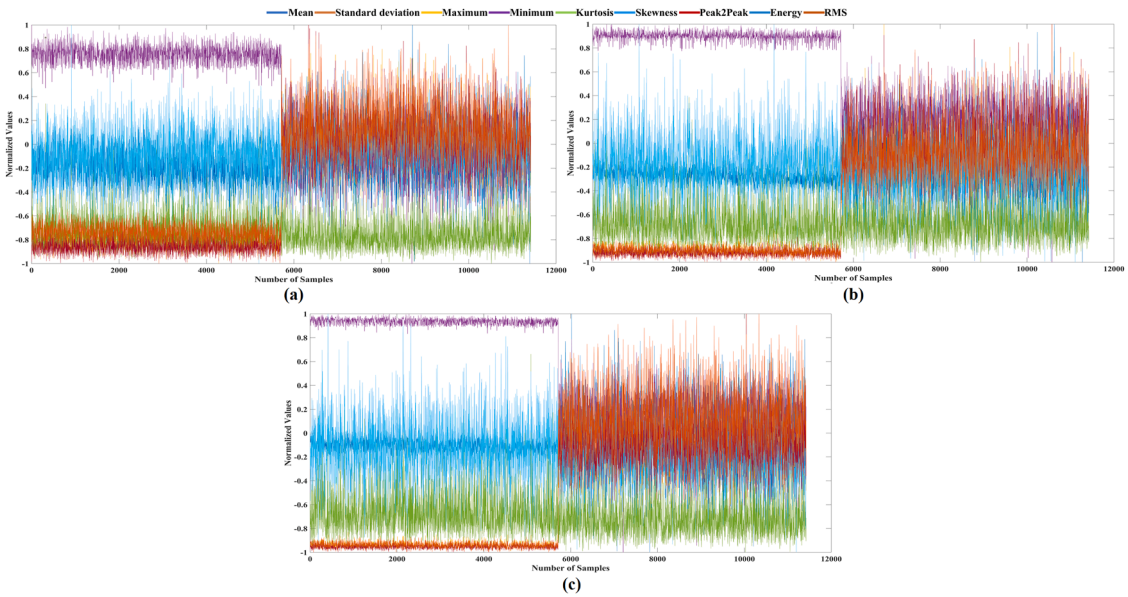


Fig. 6. Feature visualization for the GTR dataset (a) 19 Hz (b) 23 Hz (c) 29 Hz.

perfect i.e., they can distinguish the gear conditions with an accuracy of 100 %. However, the detailed results of the SVM are only considered here. The results in terms of confusion matrices are given in Table 4. Also, the six performance measures are provided in Table 5. It can be observed that all have a value of 1, as the distinction between the two classes is perfect in all speed conditions.

5.2. Classification results for the BTR dataset

In fault detection of bearing, a multi-class classification problem is considered. A total of three conditions are investigated, including healthy, inner race fault, and ball fault. The feature visualization at all the three-speed conditions is shown in Fig. 7. It is observed that most of the features have a significant margin in the three conditions. However, some features have a narrow margin in BHL and BIR conditions. In such cases, classifiers must work in higher dimensions to distinguish between the different classes, and SVM is considered efficient in such situations.

Table 6 shows the classification accuracies of different classifiers at different speed conditions. SVM performed best, with the highest accuracy of 99.3 % at 19 Hz. Another important observation is that the accuracy decreases as the speed increases. However, the accuracy is above 90% at all the speed conditions in the case of SVM, which can be considered substantial when dealing with MEMS-based economical sensors.

The detailed results for bearing dataset in terms of confusion matrix using SVM are given in Table 7. The table shows a slight confusion between BHL and BIR conditions at 19 Hz and 23 Hz, but confusion gets shifted between BIR and BBF at 29 Hz. The reason for this is that at a lower speed, intermittent vibrations are generated due to ball defect, but at a higher speed, its consistent contact with the inner and outer race produces continuous vibrations. Further, the performance measures at all three-speed conditions are given in Table 8. It can be observed that precision, recall, F1-score, specificity, and MCC have a value of more than 0.9 at 19 Hz and 23 Hz. At 29 Hz speed, only MCC and kappa values are less than 0.9, which shows a slightly weak closeness in the classified instances matched with the actual fault condition of the bearing.

5.3. Classification results for the IM dataset

As described earlier, induction motors are considered a prime source of mechanical energy transfer to rotating machines, and the timely detection of anomalies in these machines is an urgent need. In the present case, a multi-class classification problem is considered for induction motors, including MHL, MBR, and MML, as described in Section 4. The feature visualization of induction motor data in Fig. 8 reveals the margin clarity in a significantly powerful way, with some having weak overlap in the case of healthy and misaligned conditions. However, in the case of misaligned conditions, the scattering is more in range compared to healthy conditions. In Table 9, the performance comparison shows that SVM has more, or the same performance compared to other classification models.

The confusion matrices based on the results obtained through SVM are given in Table 10. Here, it is observed that the misclassification is significantly less at 19 Hz and 23 Hz, but at 29 Hz, the classifier is confused between healthy and misaligned conditions. This could be due to the mitigation of the misalignment with the dominance of vibration due to other components. Table 11 shows the results of performance measures, and it can be observed that at 19 Hz and 23 Hz, all have values more than 0.9. However, at 29 Hz, the value for all measures got a dip, but still, it is more than 0.9, except for the kappa value, which has a value lower than 0.9.

5.4. AIoT-based monitoring and real-time anomaly detection

The results from the classification part reveal that among the four classification models, the performance of SVM and RNN is best for multi-class classification. However, for binary classification, i.e., in the case of the gear dataset, all classifiers provided 100 % accuracy. In the case of model deployment to the edge device and real-time fault detection to the AIoT platform, SVM seems to be a preferred choice as it can work well with less computational burden. However, RNN can also be deployed, but the computational burden will be higher in that case. So, based on the results and its inherent advantages, SVM is deployed to the edge device for anomaly detection. For real-time detection of different fault conditions at the AIoT platform, data acquisition is started through the Raspberry Pi, and random tests are performed on the three rotating machines. The feature extraction and processing are done the same way as during the training and testing. The observations of health indicators are also simultaneously done and matched with the previously calculated values. In the case of GTR for three speeds, all conditions are predicted and updated correctly on the AIoT platform through Wi-Fi, with an average elapsed time of 0.39 s. Further, in the case of multi-class fault detection on BTR and IM, 27 random tests are done at all three speeds and repeated three times on each condition, as given in Table 12. These random tests are done as there is some misclassification in the case of 23 Hz and 29 Hz speeds. It is observed that at 19 Hz, the detection is accurate in all the conditions of BTR and IM, with an average elapsed time of 0.65 s and 0.59 s, respectively.

However, in the case of BTR, at 23 Hz, one condition, and at 29 Hz, two conditions are wrongly detected, as given in Table 12.

Table 3
Performance evaluation of various classifiers for the GTR dataset.

Speed	NB	k-NN	RNN	SVM
19 Hz	100	100	100	100
23 Hz	100	100	100	100
29 Hz	100	100	100	100

Table 4
Confusion matrices of the GTR dataset based on SVM.

True Class	Predicted Class					
	19 Hz		23 Hz		29 Hz	
	GHL	GFG	GHL	GFG	GHL	GFG
GHL	1142	0	1142	0	1142	0
GFG	0	1142	0	1142	0	1142

Table 5
Performance measures for the GTR dataset based on SVM.

Speed	Precision	Recall	F1- Score	Specificity	MCC	Kappa value
19 Hz	1.00	1.00	1.00	1.00	1.00	1.00
23 Hz	1.00	1.00	1.00	1.00	1.00	1.00
29 Hz	1.00	1.00	1.00	1.00	1.00	1.00

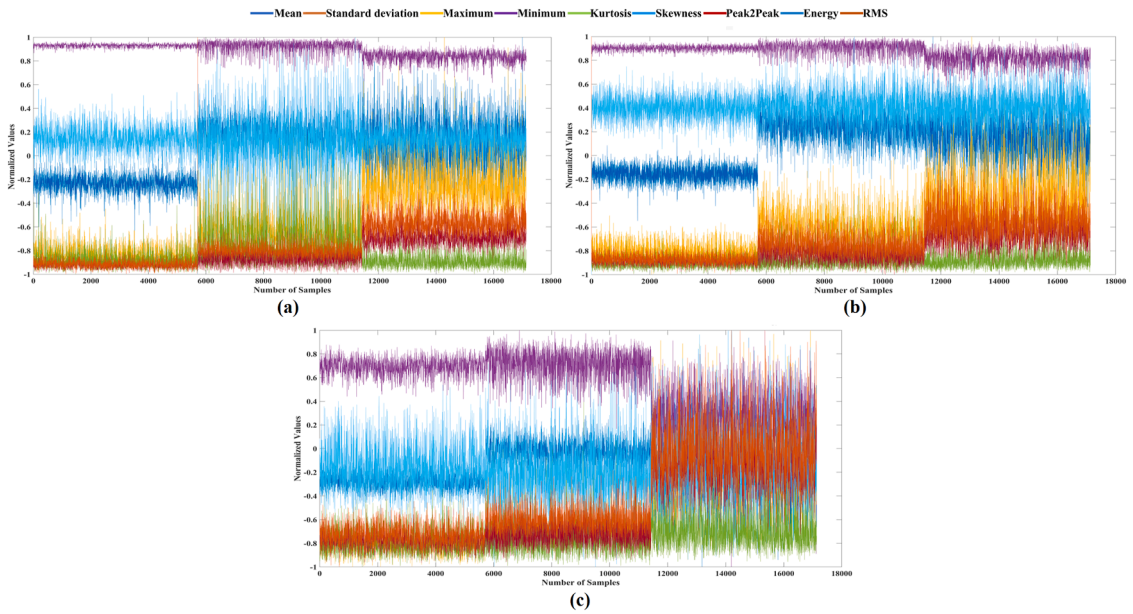


Fig. 7. Feature visualization for the BTR dataset (a) 19 Hz (b) 23 Hz (c) 29 Hz.

Table 6
Performance evaluation of various classifiers for the BTR dataset.

Speed	NB	k-NN	RNN	SVM
19 Hz	97.63	99.03	99.12	99.33
23 Hz	93.54	95.32	96.44	96.62
29 Hz	88.91	88.71	90.11	91.02

Table 7
Confusion matrices of the BTR dataset based on SVM.

True Class	Predicted Class								
	19 Hz			23 Hz			29 Hz		
	BHL	BIR	BBF	BHL	BIR	BBF	BHL	BIR	BBF
BHL	1134	8	0	1120	22	0	1136	6	0
BIR	5	1134	3	53	1079	10	8	1001	133
BBF	0	9	1133	1	31	1110	9	151	982

Table 8
Performance measures for the BTR dataset based on SVM.

Speed	Precision	Recall	F1- Score	Specificity	MCC	Kappa value
19 Hz	0.9927	0.9927	0.9927	0.9964	0.9891	0.9836
23 Hz	0.9661	0.9658	0.9659	0.9829	0.9489	0.9232
29 Hz	0.9101	0.9104	0.9102	0.9552	0.8655	0.7984

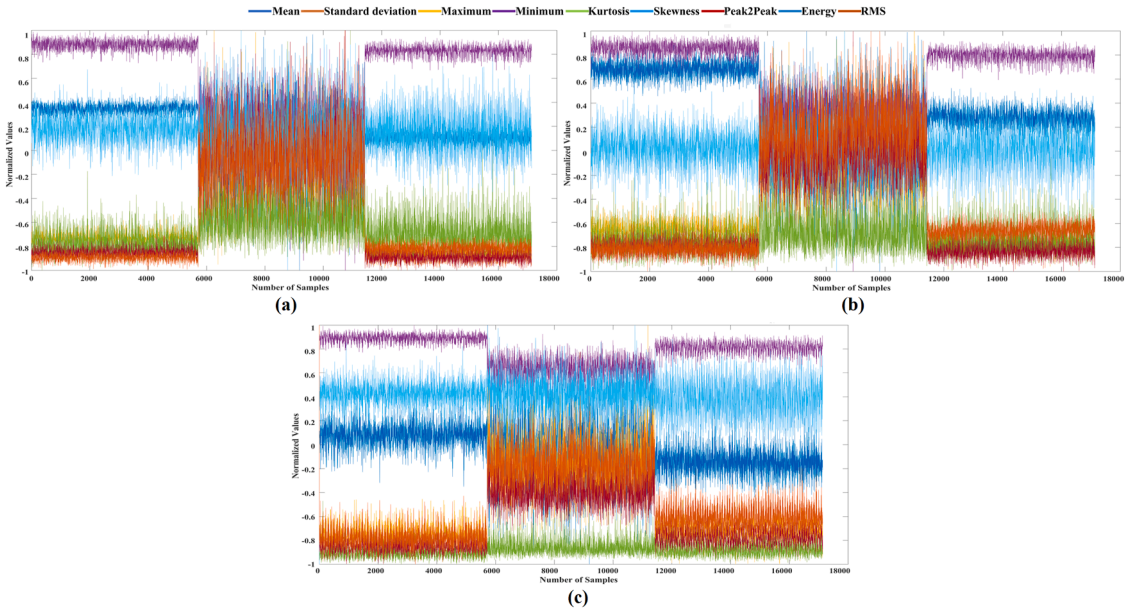


Fig. 8. Feature visualization for IM dataset (a) 19 Hz (b) 23 Hz (c) 29 Hz.

Table 9
Performance evaluation of various classifiers for the IM dataset.

Speed	NB	k-NN	RNN	SVM
19 Hz	99.03	99.36	99.85	99.92
23 Hz	99.12	99.19	99.92	99.86
29 Hz	91.95	91.37	93.58	93.64

Table 10
Confusion matrices of the IM dataset based on SVM.

True Class	Predicted Class								
	19 Hz			23 Hz			29 Hz		
	MHL	MBR	MML	MHL	MBR	MML	MHL	MBR	MML
MHL	1142	0	0	1140	0	2	1043	0	99
MBR	0	1141	1	0	1142	0	0	1133	9
MML	2	0	1140	4	0	1138	89	5	1048

Table 11
Performance measures for the IM dataset based on SVM.

Speed	Precision	Recall	F1- Score	Specificity	MCC	Kappa value
19 Hz	0.9991	0.9991	0.9991	0.9996	0.9987	0.9980
23 Hz	0.9982	0.9982	0.9982	0.9991	0.9974	0.9961
29 Hz	0.9412	0.9410	0.9411	0.9705	0.9116	0.8573

Similarly, in the case of IM, two conditions at 29 Hz are wrongly detected, as provided in Table 12. This shortcoming, i.e., slight error-prone detection for multi-class conditions at higher speed, could be due to the increased vibrations and the severity pattern of faults going close in the case of multi-class problems. In general, these results signify that the proposed system can make real-time and accurate detection of different fault conditions in the rotating machines.

5.5. Comparison of the proposed method with related work

In the literature, significantly less work has been reported on the integrated approach using edge computing, AI, and the IoT. Some researchers have done work on edge computing-based fault detection of rotating machines. Some of them used the method of data visualization on some cloud platforms. This section compares the proposed work with the latest reported work based on fault condition, used modality, classifier, maximum accuracy, and whether the AIoT-based approach has been adopted or not, and it is provided in Table 13. Tran et al. presented an intelligent method for the online monitoring of induction motor faults along with a correction scheme [21]. A 1-dimensional Convolutional Neural Network (CNN) was used for the detection of faults, and a maximum accuracy of 99.84 % was achieved. An edge computing-based solution was presented by Langarica et al. for different single-phase motor faults using a CMS2000 condition monitoring system. CNN was used for the detection of faults, and an accuracy of 95.18 % was achieved [22]. Tritschler et al. presented a strategy to diagnose bearing faults in Computer Numerical Controlled (CNC) machines using a microcontroller and a single-board computer. Frequency spectrums of different fault conditions were sent to the cloud through a message queuing telemetry transport broker [23].

Soother et al. presented an edge-based strategy for the binary classification of bearing faults using different algorithms, including Long Short-Term Memory (LSTM) network, Random Forest (RF), and SVM [24]. For the data visualization Thingspeak platform was used. The maximum accuracy achieved for the classification was 90 %. Wang et al. proposed a data reduction method for detecting bearing faults in a permanent magnet synchronous motor using vibration data and leakage flux. STM32F407 microcontroller was used for the data acquisition of different faults [25].

The proposed AIoT-based approach involves the integration of edge computing, AI, and IoT. The data is acquired and stored on the edge device with simultaneous processing and real-time fault detection through the deployed SVM model. The results on the three machines signify the robustness and reliability of the presented method for real-time anomaly detection along with remote monitoring and control.

6. Managerial and social implication

Rotating machines are integral to any moving equipment, including farm machinery, earthmovers used in mines, automotive, and wind turbines. Bearing and gears are the most crucial components in any rotating machine. Further, induction motors are widely used

Table 12
Real-time anomaly detection results on the AIoT platform for BTR and IM.

S. No	Speed (Hz)	Bearing conditions		Induction motor conditions	
		Actual condition	Predicted condition	Actual condition	Predicted condition
1		BHL	BHL	MHL	MHL
2		BHL	BHL	MHL	MHL
3		BHL	BHL	MHL	MHL
4		BIR	BIR	MBR	MBR
5	19	BIR	BIR	MBR	MBR
6		BIR	BIR	MBR	MBR
7		BBF	BBF	MML	MML
8		BBF	BBF	MML	MML
9		BBF	BBF	MML	MML
10		BHL	BIR	MHL	MHL
11		BHL	BHL	MHL	MHL
12		BHL	BHL	MHL	MHL
13		BIR	BIR	MBR	MBR
14	23	BIR	BIR	MBR	MBR
15		BIR	BIR	MBR	MBR
16		BBF	BBF	MML	MML
17		BBF	BBF	MML	MML
18		BBF	BBF	MML	MML
19		BHL	BHL	MHL	MHL
20		BHL	BHL	MHL	MML
21		BHL	BHL	MHL	MHL
22		BIR	BHL	MBR	MBR
23	29	BIR	BBF	MBR	MBR
24		BIR	BIR	MBR	MBR
25		BBF	BIR	MML	MML
26		BBF	BBF	MML	MML
27		BBF	BBF	MML	MHL

Table 13
Comparison of the proposed method with related work.

Reference (Year)	Considered faults	Modality	Classifier(s)	Maximum classification accuracy	AIoT-based monitoring and control
[21] (2023)	Induction motor faults	Vibration	1D-CNN	99.84%	No
[22] (2019)	Motor faults	Vibration	CNN	95.18%	No
[23] (2021)	Bearing faults	Vibration	–	–	No
[24] (2022)	Bearing faults	Vibration	LSTM, RF, SVM	90%	No
[25] (2021)	Bearing faults in PMSM motor	Vibration and flux	–	–	No
Proposed work	Gear faults, bearing faults, and induction motor faults	Vibration	NB, k-NN, RNN, and SVM	Binary-100% Multi-class - 99.92%	Yes

as a prime movers in rotating machines. The loss due to different faults in these components can cause the unwanted failure of the complete system and sometimes vital loss. For example, in coal mines, faults in rotating machine components cause sudden breakdowns and may even lead to unfortunate accidents. Machine-related failures account for numerous severe accidents in the mining industry.

Similarly, the bearing and gears are essential components of a wind turbine. It is quite challenging to detect any faults in an early stage due to the lack of practical availability of an effective and intelligent detection system that utilizes an integrated approach, including edge computing, AI, and IoT. In such scenarios, AIoT-based fault detection and remote monitoring could be a relevant solution for the timely detection of faults and taking necessary actions to avoid major breakdowns. The suggested AIoT-based approach can be effectively utilized for the detection of faults in different rotating machine components, including bearings, gears, and induction motors. The developed system is capable of real-time and remote monitoring of faults that helps minimize the loss by timely scheduling the maintenance. Moreover, the implementation of the suggested approach can be extended to applications related to agricultural machinery, electric vehicle motors, food processing machinery, and power plants.

7. Conclusion and future scope

Integrating AI and the IoT is very effective for real-time remote monitoring and decision-making, ensuring effective and timely predictive maintenance in rotating machines. An AIoT-based approach is proposed and implemented for real-time anomaly detection in different rotating machines, including gear test rig, bearing test rig, and induction motor. One binary classification problem related to gear faults is considered, and two multi-class classification problems are related to the bearing test rig and induction motor, respectively. The validation is presented at three different speed conditions. Four different classification models are tested, and SVM is finally adopted for deployment on the edge device based on its inherent advantage in terms of accuracy and less computational complexity. Substantially good classification results are obtained in the case of the three machines with 100 % accuracy in the binary classification of gear, and in the case of a multi-class problem of bearing, the accuracy is found to be in the range of 91.02 % - 99.33 %. This range for the multi-class classification of the induction motor is 93.64 % - 99.92 %. Further, the rotating machines' real-time anomaly detection and control are successfully demonstrated with the developed AIoT-based system. As future work, the next phase of the study will include an enhanced design with more controlling parameters for real-time remote monitoring and control of different rotating machines.

Declaration of Competing Interest

The authors have no potential conflict of interest for this work.

Data availability

Data will be made available on request.

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